

Introduction to Hypothesis Testing

- Point estimation and confidence intervals are useful statistical inference procedures.
- Another type of inference is frequently used which concerns tests of hypotheses.
- We are interested in a random variable X which has density $f(x; \theta)$ where $\theta \in \Omega$. We would like to know whether $\theta \in \omega_0$ or $\theta \in \omega_1$, where $\omega_0 \cup \omega_1 = \Omega$.

- Let μ_1 be the mean score of midterm for Purdue male students and μ_2 be the mean score for female students.
- Hypothesis testing:

$$H_0 : \mu_1 - \mu_2 \leq 0 \quad \text{vs.} \quad H_1 : \mu_1 - \mu_2 > 0$$

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{vs.} \quad H_1 : \mu_1 - \mu_2 \neq 0$$

$$H_0 : \mu_1 \leq 75 \quad \text{vs.} \quad H_1 : \mu_1 > 75$$

- Hypothesis testing:

$$H_0 : \theta \in \omega_0 \quad \text{versus} \quad H_1 : \theta \in \omega_1.$$

- The hypothesis H_0 is referred to as the **null hypothesis**, while H_1 is referred as the **alternative hypothesis**.
- Often the null hypothesis represents no change or no difference from the past, while the alternative hypothesis represents change or difference.
- The decision rule to take H_0 or H_1 is based on a sample $X_1, X_2, \dots, X_n$ from the distribution of X .
- *We may make mistakes!*

Table 1: 2×2 Decision Table for a Test of Hypothesis

Decision	H_0 is True	H_1 is True
Reject H_0	Type I Error	Correct Decision
Accept H_0	Correct Decision	Type II Error

$$\text{Type I Error} = \mathbb{P}(\text{Reject } H_0 \mid H_0 \text{ is true})$$

$$\text{Type II Error} = \mathbb{P}(\text{Accept } H_0 \mid H_1 \text{ is true})$$